

Teens' Leisure Demand: Explaining Shifts during the Smartphone Era

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Introduction

- ① Teenagers' socializing with friends is a form of **productive leisure**, particularly for social & non-cognitive skills.
- ② The early 2010s exhibited shifts in teenagers' leisure time allocation, with **in person time with friends sharply decreasing**.

This Paper

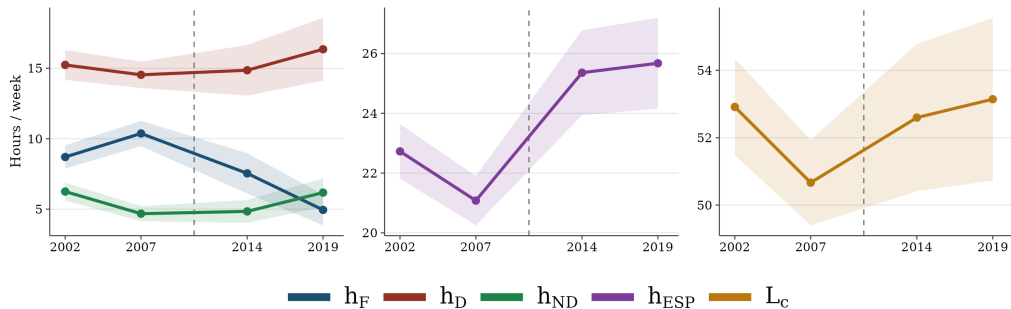
- What changes in leisure technology can rationalize this shift in time allocation?

Results

- Total teen leisure time remains largely stable. Declines in leisure time with friends are due to substitution towards other leisure activities.
- Digital technological change cannot explain the observed allocation shift. ESP technology decline is the only **statistically significant** change.

Trends in Teen's Time-Use (CDS-PSID)

Leisure Definitions: (L_c) is the sum of discretionary leisure and ESP. Discretionary leisure includes any time with friends (h_F), and residuals into digital (h_D) and non-digital (h_{ND}) buckets. Eating, sleeping and personal care (h_{ESP}) is net of 8 hours to account for night sleep.



Main Takeaway: Time with friends plummeted for 12-17 year olds ($h_F \downarrow$), shifting towards eating, sleeping, & personal care ($h_{ESP} \uparrow$) and time on digital technologies ($h_D \uparrow$). L_c Trend

Related Literature & Model Overview

Related Literature

- Measuring Leisure-Demand & Teen Leisure Trends
 - (Aguiar et al., 2021) (Kannan and Veazie, 2023) (Mollborn et al., 2022) (Zablotsky, 2024)

Contribution: Identifies what drove the change in Teens' Leisure Allocation

Model Overview:

- Teen (k) takes total leisure time (L_c) as given, and allocate leisure across leisure activities ($h_{j,k}$, $j \in \{F, D, ND, ESP\}$)
- Each teen faces period specific leisure technology $\psi_{j,t}$ and idiosyncratic leisure activity tastes $\xi_{j,k}$, where $\mathbb{E}[\log(\xi_{j,k})] = 0$.
- Teens solve a static leisure demand problem. They are myopic in how leisure allocation affects future skill generation.

Teen's Leisure Allocation Problem¹

Teen's Problem:

$$\begin{aligned} v(L_c; \Psi_t, \Xi_k) = \max \sum_j \frac{(\psi_{j,t} \xi_{j,k} h_{j,k})^{1 - \frac{1}{\eta_j}}}{1 - \frac{1}{\eta_j}} \\ \text{s.t. } \sum_j h_{j,k} \leq L_c \quad (\omega) \end{aligned}$$

Optimality Conditions:

$$h_{j,k} = (\psi_{j,t} \xi_{j,k})^{\eta_j - 1} \omega^{-\eta_j} \quad (1)$$

$$\sum_j (\psi_{j,t} \xi_{j,k})^{\eta_j - 1} \omega^{-\eta_j} = L_c \quad (2)$$

¹The leisure demand model is based off of “Leisure Luxuries and the Labor Supply of Young Men”, Aguiar, Bils, Charles and Hurst (ABCH) (2021 JPE)

Elasticities

These equations lead to the following elasticities:

- Changes w.r.t total leisure time (L_c) Derivation

$$\beta_j \equiv \left. \frac{d \log h_j}{d \log L_c} \right|_{\psi} = \frac{\eta_j}{\bar{\eta}} \quad (3)$$

where $\bar{\eta} \equiv \sum_j s_j \eta_j$ and $s_j \equiv \frac{h_j}{L_c}$. Activity h_j is a "leisure luxury" if $\beta_j > 1$ and "leisure necessity" if $\beta_j < 1$.

- Change w.r.t leisure technological change (ψ_j)² Derivation

$$\frac{d \log h_j}{d \log \psi_{-j}} = -(\eta_{-j} - 1)\beta_j s_{-j} \quad (4)$$

$$\frac{d \log h_j}{d \log \psi_j} = (\eta_j - 1)(1 - s_j \beta_j) \quad (5)$$

²An adding up condition gives $\sum_j s_j \beta_j = 1$, and thus the sign of the technology elasticity depends on if $\eta_j \leq 1$.

Comparative Statics

- From the data, L_c remains stable. Thus leisure technology change drives changes in the leisure allocation.

$\psi_j \uparrow$	h_F	h_D	h_{ND}	h_{ESP}
ψ_F ($\hat{\eta}_F = 0.90$)	-0.08	+0.03	+0.02	+0.01
ψ_D ($\hat{\eta}_D = 1.91$)	-0.20	+0.49	-0.34	-0.17
ψ_{ND} ($\hat{\eta}_{ND} = 1.58$)	-0.05	-0.10	+0.50	-0.04
ψ_{ESP} ($\hat{\eta}_{ESP} = 0.76$)	+0.08	+0.16	+0.13	-0.18

Notes: Entry = $d \log h_i / d \log \psi_j$, evaluated at $\hat{\beta}_j = \hat{\eta}_j / \bar{\eta}$, $s_j = h_j / L_c$ from 2002/2007 weighted-avg. shares ($s_F=.19$, $s_D=.29$, $s_{ND}=.11$, $s_{ESP}=.42$; $\bar{\eta}=1.20$).

- Recall $h_F \downarrow$, $h_D \uparrow$, $h_{ESP} \uparrow$, while h_{ND} is stable $\Rightarrow$ **no single technological change can explain the observed data.**
- With $\hat{\eta}_F$ close to one, rationalizing h_F 's decline requires a large shift in friend-time technology.

Data

- Teen time-use data is from the 2002, 2007, 2014, and 2019 waves of the Child Development Supplement (CDS) of the PSID.
- Total leisure time:

$$L_c = 24 - \underbrace{8}_{\text{ESP floor eat/sleep/hygiene}} - \underbrace{h_{\text{school}}}_{\text{school + homework}} - \underbrace{h_X}_{\text{extra-curriculars}} - \underbrace{\tau}_{\text{parent-active leisure time}} - \underbrace{h_{\text{other}}}_{\text{chores, work, errands, travel}}$$

Regression Model

- Budget-share AIDS regression (Deaton and Muellbauer, 1980) $\Rightarrow \beta_j = 1 + \frac{\gamma_j}{s_j}$:

$$s_{jkt} = \delta_t + \alpha_{n(k)} + \gamma_j \log L_c + u_{jkt}$$

- **Exclusion restriction** $u_{jkt} \perp L_c$. Bias on $\hat{\beta}_D \uparrow$ if taste for video games leads to shirking from homework or if teen studies late at night because they are an efficient sleeper $\beta_{ESP} \downarrow$

Elasticity Results

- **Setup:** Diaries are grouped into synthetic cells (wave $\times$ age $\times$ sex $\times$ race $\times$ region) with $n \geq 5$ cutoff to address zeros $\rightarrow$ 2,888 time diaries and 250 synthetic cells.

Table: Leisure Engel Curves of Teens 12–17: $\hat{\beta}_j$

	(1) Wave	(2) +Age	(3) +Sex	(4) +Race	(5) +Region
$\hat{\beta}_F$	0.62 (0.21)	0.74 (0.21)	0.65 (0.21)	0.75 (0.21)	0.76 (0.21)
$\hat{\beta}_D$	1.78 (0.14)	1.81 (0.15)	1.64 (0.14)	1.59 (0.14)	1.62 (0.14)
$\hat{\beta}_{ND}$	1.08 (0.24)	1.02 (0.25)	1.16 (0.25)	1.31 (0.26)	1.29 (0.26)
$\hat{\beta}_{ESP}$	0.61 (0.09)	0.56 (0.09)	0.67 (0.09)	0.63 (0.09)	0.61 (0.09)

Fixed Effects: column headers show the FE cumulatively added at each step. *Notes:* Cells are weighted by the sum of their members' survey weights. Bootstrap standard errors in parentheses (1,000 parent-clustered resamples, cells rebuilt each draw). Column 4 is the chosen benchmark.

Identifying Technological Change

Structural Elasticity Estimates

- With elasticity estimates, $\beta_j = \frac{\eta_j}{\bar{\eta}}$, there are 4 equations and 5 unknowns.
- Assuming ESP elasticities are common across populations, we anchor η_{ESP} to Aguiar et al. (2021), using an IVW estimate over the four age/gender ESP elasticities. ABCH Anchor
- Results: $\hat{\eta}_{ESP} = 0.76$ (SE 0.07) $\Rightarrow \hat{\eta}_F = 0.90$ (SE 0.36), $\hat{\eta}_D = 1.91$ (SE 0.37), $\hat{\eta}_{ND} = 1.58$ (SE 0.39).

Technological Change

- Identifying technology change requires a reference activity that is assumed to be unchanged across periods. The chosen reference is non-digital leisure $\log(\psi_{ND,t}) \equiv 0 \forall t$.
- Optimality Conditions normalizing $\mathbb{E}[\log \xi_{j,k}] = 0$:

$$\underbrace{\log(\xi_{j,k}) - \frac{\eta_j(\eta_{ND} - 1)}{\eta_{ND}(\eta_j - 1)} \log(\xi_{ND,k})}_{\mathbb{E}[\cdot] = 0} = \frac{1}{\eta_j - 1} \log(h_{j,k}) - \frac{\eta_j}{\eta_{ND}(\eta_j - 1)} \log(h_{ND,k}) - \log(\psi_j)$$

Technological Change Results

- Thus the technology change comes from the sample mean:

$$\widehat{\log(\psi_j)} = \frac{1}{N} \sum_{k=1}^N \left[\frac{1}{\eta_j - 1} \log(h_{j,k}) - \frac{\eta_j}{\eta_{ND}(\eta_j - 1)} \log(h_{ND,k}) \right]$$

	$\log \psi_{j,t_0}$	$\Delta \log \psi_{j,2014}$	$\Delta \log \psi_{j,2019}$
ψ_F	-13.00 [-25.71, 26.95]	+5.07 [-17.93, 14.89]	+9.32 [-34.61, 29.07]
ψ_D	0.87 [-2.69, 8.69]	+0.10 [-0.74, 2.27]	+0.29 [-1.23, 1.84]
ψ_{ESP}	-9.75 [-22.84, -4.68]	-1.18 [-3.59, -0.33]	-1.19 [-3.34, -0.17]
N cells	168	37	41
N diaries	2,215	310	339

Notes: $t_0 = 2002-07$ pooled; Δ vs. t_0 . Brackets are 95% percentile CIs (family-clustered bootstrap). 4/250 cells dropped for zeros (3 are $h_F=0$, in 2014/2019) $\Rightarrow \psi_F$'s change is biased.

- h_D 's technology remains insignificant; h_F 's CI spans zero in both periods; h_{ESP} 's technology decline is significant in both periods.

No evidence was found for digital technology improvement $\Delta\psi_{D,t}$.

- Time diary methodology has issues identifying the ubiquity of phones in everyday life, likely understating time-use on these devices.

ESP technology sees a rapid decline over this period.

- This does not mean these basic activities saw technological decline, but that more time is potentially spent on them when phone use is a secondary activity. [Further Discussion](#)

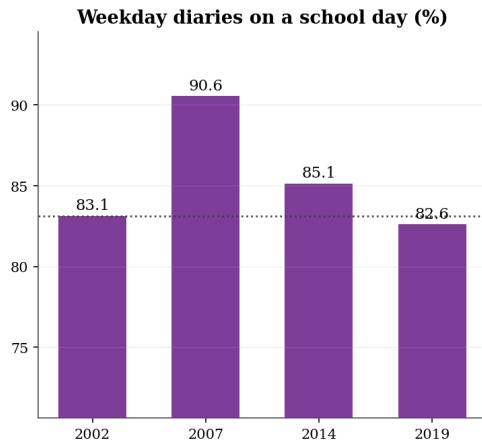
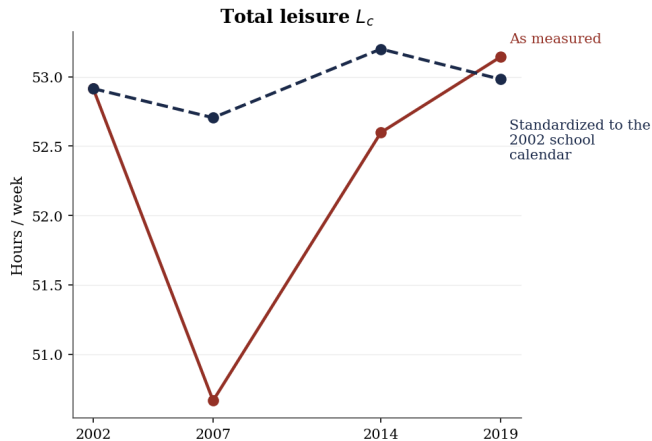
Friend-time technology is a noisily estimated; 2019 CI [-34.61, 29.07].

- $\hat{\eta}_F = 0.90$ (SE 0.36) is the one estimate well within the $\eta_j \gtrless 1$ threshold where the technological sign story flips.

Next Steps

- Time a teen spends socializing is a productive form of leisure, improving social and non-cognitive skills.
- From the perspective of parents, a decline in returns to teen leisure would imply increasing time and money investment $\tau \uparrow, h_X \uparrow$.
- How can an increased incentive to invest in your child be reconciled with L_c being flat across this time period?
 - Increasing value of parental leisure? Total leisure remains flat from (2004-2025).
 - Economic conditions driving parents to spend more time in the labor market? Mothers $\uparrow\uparrow$, Fathers $\downarrow$, Net $\uparrow$ between (2004-2025).
 - Screen use as a substitute for child-care?

Total Leisure Time Trends



- **Main Takeaway:** Re-weighting diaries to reflect the 2002's fraction of weekday diaries where the teen is in school leads to L_c being stable over time.

Total Leisure Change Elasticity Derivation

- Holding fixed the technology vector ψ and using the FOC w.r.t activity h_j , we have:

$$\log h_j = (\eta_j - 1) \log \psi_j - \eta_j \log \omega \Rightarrow \frac{d \log h_j}{d \log \omega} = -\eta_j$$

- Using the FOC summation over the time constraint:

$$\frac{\partial}{\partial L_c} \left[\sum_j \psi_j^{\eta_j-1} \omega^{-\eta_j} - L_c \right] = 0 \Rightarrow \frac{d \log \omega}{d \log L_c} = - \frac{1}{\sum_j \eta_j h_j \omega^{-1}} \frac{L_c}{\omega}$$

- Combining gives: [◀ Back to Main Slide](#)

$$\frac{d \log h_j}{d \log L_c} = \frac{d \log h_j}{d \log \omega} \frac{d \log \omega}{d \log L_c} = \frac{\eta_j}{\sum_j \eta_j \frac{h_j}{L_c}}$$

Technological Change Elasticity Derivation

- Holding fixed L_c and using the FOC w.r.t. activity h_j , totally differentiating:

$$d \log h_j = (\eta_j - 1) d \log \psi_j - \eta_j d \log \omega$$

- Implicit differentiation over equation (2) gives:

$$(\eta_j - 1) \psi_j^{\eta_j - 2} \omega^{-\eta_j} + \frac{d\omega}{d\psi_j} \sum_j (-\eta_j) \psi_j^{\eta_j - 1} \omega^{-\eta_j - 1} = 0 \Rightarrow$$

$$\frac{d \log \omega}{d \log \psi_j} = \frac{(\eta_j - 1) h_j}{\sum_j \eta_j h_j} \Rightarrow \frac{(\eta_j - 1) s_j}{\bar{\eta}} \Rightarrow$$

$$\frac{d \log h_j}{d \log \psi_j} = \underbrace{(\eta_j - 1)}_{\text{direct effect}} - \underbrace{\beta_j s_j (\eta_j - 1)}_{\text{indirect, via } \omega}$$

ABCH Anchor: Additional Detail

- (Aguiar et al., 2021) use ATUS time diaries and an AIDS regression, as well as an external anchor, to identify η_j .
- ESP is defined nearly identically, with the caveat that our ESP is net of 8 hours, not 7, to reflect different sleep needs.

	$\hat{\beta}_{ESP}$ (SE)	Cells	Individuals
Younger Men	0.76 (0.10)	281	6,780
Older Men	0.61 (0.04)	509	36,715
Younger Women	0.64 (0.09)	275	10,246
Older Women	0.65 (0.04)	463	44,149

- All four $\hat{\beta}_{ESP}$ results are statistically indistinguishable; we pool them via inverse-variance weighting around a common external anchor $\bar{\eta} = 1.19$ (SE 0.10):

$$\hat{\eta}_{ESP} = \left(\sum_g \widehat{\text{Var}}(\hat{\eta}_{ESP,g})^{-1} \right)^{-1} \sum_g \widehat{\text{Var}}(\hat{\eta}_{ESP,g})^{-1} \hat{\eta}_{ESP,g}$$

- Result: $\hat{\eta}_{ESP} = 0.76$ (SE 0.07). [◀ Back](#)

Largest ESP changes (3 groups)

	2002	2019	Δ (SE)
Night sleep	63.68	65.21	1.53** (0.77)
Meals & snacks	6.48	7.17	0.69*** (0.25)
Hygiene/grooming	6.24	6.90	0.66** (0.27)

Notes: Weighted avg. weekly hrs, CDS teens 12–17. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Secondary activity during meals/snacks: None reported for sleep or hygiene.

	2002	2007	2014	2019
None coded	34.2%	28.7%	44.3%	51.7%
Conversation/social	43.8%	43.5%	25.6%	17.6%
TV	16.5%	20.4%	18.6%	20.6%
Digital/games/music/phone	1.9%	4.2%	9.8%	8.6%
Other (chores/school)	3.5%	3.3%	1.7%	1.6%
<i>N</i> spells	4,121	4,358	1,894	1,964
<i>N</i> children	709	699	285	283

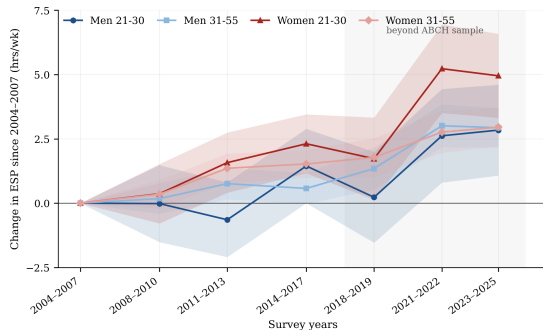
Notes: Wtd. share of meal/snack (h_{ESP}) diary time by coded secondary (“what else”) activity, CDS teens 12–17. “None coded” = no secondary activity recorded, not necessarily none occurring.

▶ Continued

◀ Back to Discussion

ESP Discussion (Cont.)

- However, outside research explains this phenomenon in teens for sleep, in the bathroom, and while eating meals.³



- **Main Takeaway:** All 4 gender/age groups also see an ESP rise, although lagged. Note discretionary leisure time remains largely stable across this time-period. [◀ Back](#)

³Sleep: Brosnan et al. (2024). Bathroom: Ramprasad et al. (2025) Meals: Garg et al. (2025)

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